

Demystifying DNS: How Websites Load in Seconds

DNS is the phone book of the internet, turning human-friendly names into computer-friendly numbers. Understanding how it works is essential for troubleshooting connection issues in your daily engineering work.

What you'll learn:

- 1 Checking Your Local DNS Cache
- 2 Querying DNS with Nslookup
- 3 Tracing the Full DNS Path with Dig
- 4 Clearing Bad DNS Records

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Checking Your Local DNS Cache

Before your computer asks the internet for a website address, it checks its own memory first. If you visited a site recently, the address is saved right on your machine. You can view these saved addresses instantly using your command prompt.

```
$ipconfig /displaydns
```

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Querying DNS with Nslookup

When you want to manually ask a DNS server for a website's IP address, nslookup is your best friend. It lets you query specific domain names and see exactly what numbers they return. This is the first tool senior engineers use during network outages.

```
$nslookup github.com
```

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Tracing the Full DNS Path with Dig

The dig command gives you a deep, detailed look into how the internet resolves a domain name. By using the trace flag, you can watch your request hop from root servers all the way to the final authoritative server.

```
$dig +trace amazon.com
```

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Clearing Bad DNS Records

Sometimes DNS records update, but your computer holds onto the old address causing connection errors. Flushing your local DNS cache forces your operating system to forget old entries and fetch fresh data from the network.

```
$ipconfig /flushdns
```


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Key Takeaways

- + Always test DNS resolution from multiple networks if a site fails
- + Keep the dig command handy for detailed debugging sessions
- + Use public DNS providers like 1.1.1.1 or 8.8.8.8 for reliable lookups
- + Remember that DNS changes can take time to spread across the globe
- + Document your network troubleshooting steps for your team

Watch Out For

- ! Assuming a website is down globally when it is just a local cache issue
- ! Forgetting to flush your DNS cache after updating domain records
- ! Confusing browser cache with operating system DNS cache

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